Learning Expressive Latent Priors via Latent Diffusion Models in Variational Autoencoders

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Abstract

This paper investigates a two-stage generative modeling framework designed to mitigate the inherent trade-offs between Variational Autoencoders (VAEs) and diffusion-based models. While VAEs offer efficient, single-pass inference, they frequently produce blurry samples due to the information bottleneck imposed by oversimplified Gaussian priors. Conversely, although diffusion models achieve state-of-the-art image quality, their iterative sampling in high-dimensional pixel space remains computationally prohibitive.

We propose a decoupled approach that separates representation learning from prior generation. In the first stage, we employ a hierarchical Nouveau VAE (NVAE) as a "compressor" to map high-resolution images into a structured latent space. In the second stage, we replace the standard normal prior with an expressive Inductive Moment Matching (IMM) prior trained via flow-matching on interpolated paths. Unlike traditional distillation, IMM provides a single-stage training procedure that guarantees distribution-level convergence and enables high-fidelity, few-step sampling.

Experimental results on the CelebA dataset demonstrate that our NVAE-IMM framework significantly enhances generative fidelity, achieving a state-of-the-art Fréchet Inception Distance (FID) of 33.35 with only 16 to 64 sampling steps. This work establishes a robust foundation for hybrid latent diffusion architectures that balance probabilistic reconstruction with high-speed, structured image synthesis.

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Learning latent representations*; *Generative and developmental approaches.*

Keywords

variational autoencoders, diffusion models, latent priors, NVAE, IMM, DDPM, generative modeling

ACM Reference Format:
Muhammad Imran Hossain. 2025. Learning Expressive Latent Priors via Latent Diffusion Models in Variational Autoencoders. In *CS 793A: Generative AI Project, Spring 2025, Morgantown, WV, USA*. ACM, New York, NY, USA, 6 pages. https://doi.org/10.0000/0000000.0000000

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CS 793A '25, Morgantown, WV, USA
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ACM ISBN 000-0-0000-0000-0
https://doi.org/10.0000/0000000.0000000

1 Introduction

Deep generative models have advanced significantly, with Variational Autoencoders (VAEs) [6], Generative Adversarial Networks (GANs), and diffusion models [4] emerging as the dominant paradigms. VAEs provide a principled probabilistic framework with explicit latent variables and tractable training via the Evidence Lower Bound (ELBO) [6]. However, they frequently produce blurry samples due to the "prior mismatch" problem, where the true data distribution is forced into an oversimplified standard Gaussian prior. Conversely, Denoising Diffusion Probabilistic Models (DDPMs) have established new benchmarks in image quality but remain computationally expensive because they operate directly in high-dimensional pixel space, requiring hundreds to thousands of iterative denoising steps [4].

This research is motivated by the inherent trade-offs between these two architectures:

- **VAEs are efficient but lack fidelity:** While sampling requires only a single pass through the decoder, the Gaussian prior forms a bottleneck that fails to capture the complex, multimodal structure of real latent codes.
- **Diffusion models are expressive but slow:** These models generate high-fidelity images but suffer from high computational costs during inference due to their iterative nature in pixel space [4].

To address these limitations, we adopt a *Latent Diffusion* approach [12]. By first learning a compact latent representation via a hierarchical VAE (NVAE-style) [13], we can perform diffusion-based generative modeling within a lower-dimensional manifold. This project instantiates this idea by using an NVAE as a hierarchical "compressor" and learning expressive priors over the global latent space. Specifically, we investigate the recently proposed Inductive Moment Matching (IMM) framework [16], which approximates a transport map from Gaussian noise to the empirical latent distribution via an Ordinary Differential Equation (ODE) flow, enabling significantly faster sampling than traditional DDPMs [16].

The primary contributions of this project are

- A two-stage architecture that decouples representation learning (Stage 1) from prior learning (Stage 2) in latent space.
- A hierarchical latent design separating global vectors and spatial tensors, which substantially improves reconstruction quality over single-layer VAEs on the CelebA dataset.
- A systematic comparison of Gaussian, DDPM, and IMM priors using Fréchet Inception Distance (FID), MSE/LPIPS, and t-SNE visualizations. Our results demonstrate that an NVAE combined with an IMM prior achieves the optimal balance between image sharpness and sampling efficiency.

2 Background

2.1 Variational Autoencoders (VAEs)

Variational Autoencoders (VAEs) provide a probabilistic framework for learning latent representations of complex data. A VAE consists of an encoder $q_\phi(z|x)$ that maps input data x to a latent variable z, and a decoder $p_\theta(x|z)$ that reconstructs the data from the latent space [6]. The model is trained by maximizing the Evidence Lower Bound (ELBO):

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - D_{KL}(q_\phi(z|x)||p(z)) \quad (1)$$

The first term encourages high reconstruction fidelity, while the second term regularizes the approximate posterior $q_\phi(z|x)$ towards a fixed prior $p(z)$, typically a standard Gaussian $\mathcal{N}(0, I)$.

2.2 Hierarchical VAEs (NVAE)

While standard VAEs are computationally efficient, they often suffer from blurry generations due to the restrictive assumption that the complex posterior distribution can be mapped to a simple Gaussian. To address this, Hierarchical VAEs such as the Nouveau VAE (NVAE) [13] and very deep VAEs [1] introduce a multi-scale latent structure. In these architectures, the latent space is split into groups $\{z_1, z_2, \ldots, z_L\}$, allowing the model to separate high-level semantic information (global latents) from fine-grained spatial details (local latents). This hierarchy significantly improves the reconstruction quality and captures long-range correlations more effectively than single-layer models.

2.3 Denoising Diffusion Probabilistic Models (DDPM)

Diffusion models have emerged as a powerful alternative for high-fidelity image synthesis [4]. They operate by defining a forward process that gradually adds Gaussian noise to the data until it becomes pure noise, and learning a reverse process (parameterized by a neural network) to denoise the data step-by-step. While DDPMs achieve state-of-the-art sample quality, their inference is computationally expensive, typically requiring hundreds or thousands of iterative steps in the high-dimensional pixel space.

2.4 Inductive Moment Matching (IMM)

Inductive Moment Matching (IMM) is a recently proposed class of generative models designed to resolve the trade-off between sample quality and inference speed [16]. Unlike diffusion models that rely on long sampling chains, or distillation methods that require complex multi-stage optimization, IMM employs a single-stage training procedure to learn a transport map that aligns the moments of the noise distribution with those of the data distribution.

Formally, IMM learns a vector field $v_\theta(z, t)$ that minimizes a conditional flow-matching objective. It guarantees distribution-level convergence and is specifically optimized for one- or few-step sampling (e.g., 8 steps) without the instability often associated with distilling diffusion models. Zhou et al. demonstrate that IMM can achieve competitive performance (e.g., FID of 1.99 on ImageNet) with significantly reduced computational cost compared to standard diffusion [16].

3 Related Work

3.1 Latent Space Generative Models

To mitigate the high computational cost of pixel-space diffusion, recent works have proposed shifting the generative process to the latent space of a pre-trained autoencoder. *Latent Diffusion Models* (LDMs) [12] train a diffusion prior on the compressed representations of a VQ-GAN or VAE, effectively separating the learning of perceptual compression from semantic generation. Similarly, *Latent Score-Based Generative Models* (LSGM) [14] train a continuous-time score-based model in the latent space of a hierarchical VAE (NVAE), allowing for end-to-end optimization of both the encoder/decoder and the prior.

Our work follows this paradigm but differentiates itself by replacing the standard diffusion prior with an IMM prior. While LDMs typically rely on DDPM or DDIM samplers that still require significant steps, our approach leverages IMM to achieve rapid sampling of the latent codes. Furthermore, recent scaling efforts like SDXL [11] demonstrate that refined latent spaces are critical for high-resolution synthesis, validating our choice of using a hierarchical NVAE backbone.

3.2 Flow Matching and Fast Sampling

The inefficiency of standard diffusion sampling has catalyzed research into flow-based alternatives. *Flow Matching* [8] and *Rectified Flow* [9] generalize the diffusion process by regressing a velocity field that pushes probability paths along straight lines from noise to data. These methods reduce the curvature of the transport trajectory, enabling faster integration with ODE solvers.

Inductive Moment Matching (IMM) [16] builds upon these concepts but introduces a specific inductive bias for moment matching, enabling high performance in the few-step regime (1-8 steps). In the context of our project, IMM serves as an ideal candidate for a learned prior: it is robust enough to model the complex, potentially non-Gaussian aggregate posterior of the NVAE, yet fast enough to maintain the efficiency advantage VAEs hold over pixel-space diffusion models. By applying IMM in the latent space, we aim to combine the structural benefits of hierarchical VAEs with the generation quality and speed of modern flow-based generative models.

4 Methodology

We use a decoupled two-stage generative modeling pipeline: (i) a latent *compressor* learned via a VAE-style encoder–decoder, and (ii) an explicit *prior model* trained on the frozen latent codes for high-fidelity sampling. This separation isolates representation learning from prior expressivity and enables controlled comparisons across priors.

4.1 Stage 1: Representation Learning (Compressor)

Given images $x \in \mathcal{X}$, we learn a low-dimensional latent representation z using a variational autoencoder (VAE) with encoder $q_\phi(z|x)$ and decoder $p_\theta(x|z)$ [7]. Training maximizes the ELBO:

$$\mathcal{L}_{\text{ELBO}}(\theta, \phi) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta\, D_{KL}(q_\phi(z|x) \,\|\, p(z)), \quad (2)$$

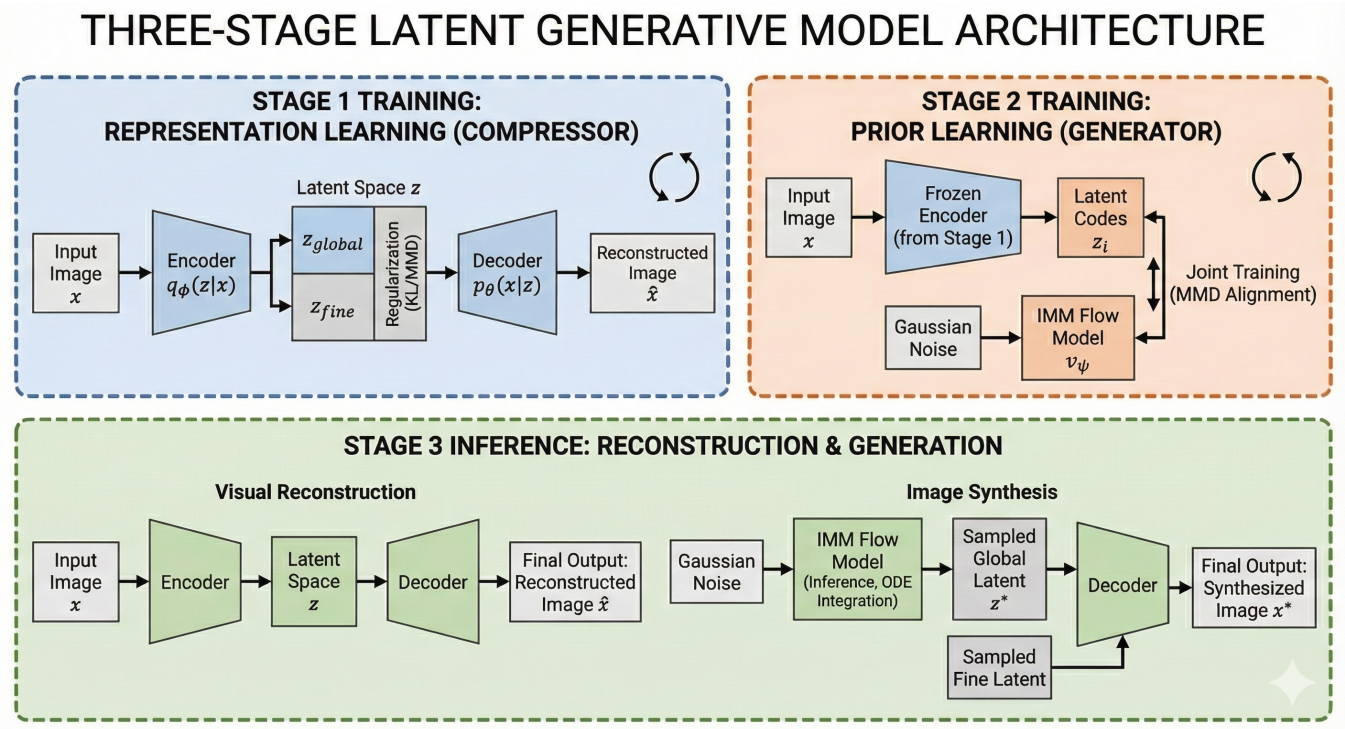


Figure 1: Overall two-stage pipeline. Stage 1 trains a VAE/NVAE compressor to map images to a regularized latent space. Stage 2 freezes the compressor, extracts latent codes, and learns an expressive latent prior (DDPM baseline vs. IMM) for generation.

where $p(z) = \mathcal{N}(0, I)$ and β controls the strength of latent regularization.

4.1.1 Architectures. We evaluate two compressor backbones:

- **Standard VAE:** a 4-layer convolutional encoder–decoder with a flat latent vector ($\dim(z) = 64$).
- **Hierarchical NVAE-style:** a multi-scale latent hierarchy inspired by NVAE [13], decomposing z into a global code z_{global} and a spatial fine-scale tensor z_{fine} to better capture both coarse semantics and high-frequency details.

4.1.2 Latent Regularization. To balance reconstruction quality and latent-space structure, we consider:

- **KL-only:** the KL term in Eq. (2).
- **MMD-only:** Maximum Mean Discrepancy to match the aggregate posterior to the prior via kernel two-sample matching [2].
- **Hybrid KL+MMD:** KL primarily regularizes z_{fine}, while MMD stabilizes z_{global}. This improves the geometry of the global latent manifold used by Stage 2.

4.2 Stage 2: Prior Learning (Generator)

After Stage 1 converges, we freeze ϕ, θ and build a latent dataset $\mathcal{D}_z = \{z^{(i)} \sim q_\phi(z|x^{(i)})\}_{i=1}^N$. We then learn a generative prior $p_\psi(z)$ to approximate $\mathcal{D}_z$.

4.2.1 Baseline Prior: Latent DDPM. As a strong latent-prior baseline, we train a denoising diffusion probabilistic model (DDPM) [5] on latent codes with the simplified noise-prediction objective:

$$\mathcal{L}_{\text{DDPM}}(\psi) = \mathbb{E}_{t,z_0,\epsilon}\left[\|\epsilon - \epsilon_\psi(\sqrt{\bar{\alpha}_t} z_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t)\|^2\right]. \quad (3)$$

4.2.2 Proposed Prior: Latent Inductive Moment Matching (IMM). To enable few-step (or single-step) sampling, we adopt Inductive Moment Matching (IMM) [16]. IMM parameterizes a time-dependent vector field $v_\psi(z_t, t)$ defining a probability path from $p_0(z)$ (Gaussian noise) to the target latent distribution. Sampling follows the ODE:

$$\frac{dz_t}{dt} = v_\psi(z_t, t), \quad t \in [0, 1]. \quad (4)$$

Training uses a Conditional Flow Matching (CFM) regression objective as instantiated in IMM [16]:

$$\mathcal{L}_{\text{IMM}}(\psi) = \mathbb{E}_{t,z_0,z_1}\left[\|v_\psi(z_t, t) - (z_1 - z_0)\|^2\right], \quad (5)$$

where $z_0 \sim \mathcal{N}(0, I)$ and $z_1 \sim \mathcal{D}_z$.

4.2.3 Stabilization: Robust Normalization and MMD Alignment. Because empirical latents may deviate from $\mathcal{N}(0, I)$, we apply:

- **Robust normalization** using encoder statistics:

$$\tilde{z} = \Sigma^{-\frac{1}{2}}(z - \mu_{\text{enc}}), \quad \mu_{\text{enc}} = \mathbb{E}_{\mathcal{D}_z}[z]. \quad (6)$$

- **MMD alignment** to discourage off-manifold latent samples:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{IMM}} + \lambda_{\text{MMD}}\,\text{MMD}^2(\{\hat{z}_i\}, \{z_{\text{vae},i}\}), \quad (7)$$

where $\{\hat{z}_i\}$ are generated latents and $\{z_{\text{vae},i}\}$ are encoder latents [2].

5 Experimental Setup

5.1 Dataset and Preprocessing

We conduct all experiments on CelebA [10]. Images are center-cropped and resized to 64×64 RGB. We use a standard train/validation/test split, where the validation set is drawn from training data for early stopping and hyperparameter selection.

5.2 Stage 1 Training (VAE/NVAE Compressor)

Unless otherwise noted, Stage 1 uses Adam with learning rate 10^{-3}, batch size 128, and up to 100 epochs. For the standard VAE, we set $\dim(z) = 64$. For the NVAE-style compressor, we use a 3-scale hierarchy with a 64-dimensional z_{global} and an $8 \times 8 \times 32$ z_{fine}.

We tune the regularization strength β (Eq. (2)) per architecture; smaller β is typically required for the hierarchical model to avoid posterior collapse [13]. We also evaluate KL-only, MMD-only, and hybrid KL+MMD regularization (Section 4).

5.3 Stage 2 Training (Latent DDPM vs. IMM Prior)

Stage 2 trains priors on the frozen *global* latents z_{global} produced by the converged NVAE encoder. The DDPM prior uses a residual MLP denoiser (hidden size 512) trained with Adam at 10^{-4} for 100 epochs using a linear noise schedule [5]. The IMM prior uses a comparable MLP backbone and is trained with the CFM/IMM objective (Eq. (5)), integrating the ODE with a small fixed number of steps at inference [16]. When enabled, Stage 2 additionally applies the normalization (Eq. (6)) and MMD alignment term (Eq. (7)).

5.4 Evaluation Protocol

We evaluate (i) compressor fidelity via reconstruction error (MSE) and perceptual similarity (LPIPS) [15], and (ii) generative quality via Fréchet Inception Distance (FID) computed over 5,000 generated samples [3]. Qualitative results are reported using matched image grids for real samples, reconstructions, and generations from each prior.

6 Results

The evaluation of our framework centers on two primary objectives: assessing the reconstruction fidelity of the compressed latent space and benchmarking the generative quality of the learned priors. We perform experiments on the CelebA dataset, center-cropped and resized to 64×64 resolution.

6.1 Reconstruction Performance

The first stage of our pipeline requires a high-fidelity compressor to ensure that semantic details are preserved in the latent space. We compared a standard 4-layer CNN VAE against a 3-scale hierarchical NVAE architecture.

As shown in Table 1, the hierarchical NVAE architecture significantly outperforms the standard VAE, achieving nearly a tenfold reduction in Mean Squared Error (MSE) and an order of magnitude improvement in perceptual similarity (LPIPS). Visual evidence of this improvement is provided in Figure 2 and Figure 3, where the NVAE reconstructions maintain significantly sharper edges and facial textures compared to the blurred outputs of the standard VAE.

Table 1: Reconstruction Fidelity Comparison

Model	MSE (↓)	RMSE (↓)	LPIPS (↓)
VAE (KL-Base)	56.06131	7.48741	0.041062
VAE (MMD-Base)	56.06073	7.487372	0.040855
NVAE (KL-Base)	**5.371127**	**2.317569**	**0.001806**
NVAE (MMD-Base)	5.291257	2.300273	0.001784

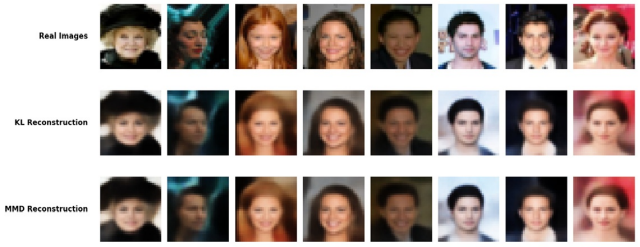


Figure 2: Reconstruction results using standard VAE.

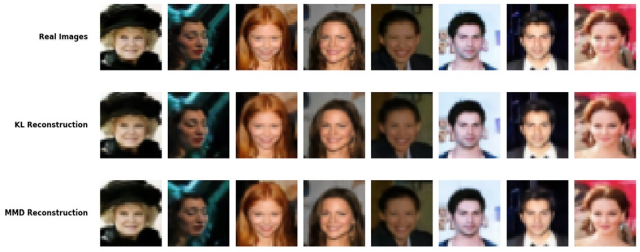


Figure 3: Reconstruction results using hierarchical NVAE.

6.2 Generative Quality and Sampling Efficiency

We benchmark the generative performance using the Fréchet Inception Distance (FID) on 5,000 samples. We evaluate two learned priors—DDPM and the proposed IMM—against the baseline Gaussian priors used in standard VAE training.

The quantitative results in Table 2 demonstrate that replacing the fixed Gaussian prior with a learned latent prior drastically improves generative quality. Most notably, the IMM prior achieves a state-of-the-art FID of 33.35 with only 64 sampling steps, outperforming the DDPM prior which requires 1,000 steps to reach an FID of 36.63. Visual samples generated by these priors are illustrated in Figure 4 and Figure 5. These grids demonstrate that the IMM prior effectively corrects the "prior mismatch" problem, yielding diverse and structured face synthesis that far exceeds the quality of samples drawn from a standard normal distribution.

Table 2: Generative Performance (FID) and Sampling Steps

Prior Method	Steps	VAE FID	NVAE FID
Standard (KL-Base)	–	147.5376	44.11937
Standard (MMD-Base)	–	50.866	42.26142
DDPM	100	38.89	37.7387
DDPM	500	37.06	36.76902
DDPM	1000	36.12	36.63936
IMM	8	36.83	34.08461
IMM	16	35.48	33.36809
IMM	32	**35.12**	33.45464
IMM	**64**	36.27	**33.35478**



Figure 4: Generative samples using VAE backbone with learned priors.



Figure 5: Generative samples using NVAE backbone with learned priors.

7 Future Work

While the proposed framework demonstrates a significant improvement in latent generative modeling, several avenues for future exploration remain:

- **Deeper Latent Hierarchies:** Current experiments utilize a 2-level NVAE structure. Scaling this to three or more hierarchical levels could enable the high-fidelity generation of 256×256 resolution images by capturing more granular spatial dependencies.
- **End-to-End Joint Tuning:** We propose fine-tuning the Stage 1 decoder simultaneously during Stage 2 prior training. Allowing loss gradients to flow back from the prior to the decoder may further align the representation and generation objectives, reducing artifacts at the boundary of the manifold.
- **Controllable Generation:** Incorporating classifier-free guidance or attribute conditioning (e.g., specific facial expressions such as "Smile") into the IMM prior would enable more precise control over the semantic output of the synthesized images.

8 Conclusions

This project investigates the synergy between hierarchical representation learning and expressive latent-diffusion priors. By decoupling the generative process into a compression stage (NVAE) and a prior learning stage (IMM/DDPM), we effectively overcome the "prior mismatch" problem inherent in standard variational autoencoders.

Our findings lead to the following key conclusions:

- **Efficiency and Expressivity:** Inductive Moment Matching (IMM) provides a computationally lightweight yet highly expressive latent prior. It achieves superior generative performance (FID) compared to standard Gaussian baselines without the prohibitive cost of pixel-space diffusion.
- **Sampling Superiority:** Quantitative benchmarks reveal that IMM achieves its best results in significantly fewer sampling steps (e.g., 16–32 steps) than traditional DDPM models, which require upwards of 1,000 steps to reach comparable fidelity.
- **Structural Fidelity:** The use of an NVAE backbone ensures that sharp reconstructions are maintained, providing a robust foundation for the learned prior to synthesize structured and realistic face images.

In summary, the integration of IMM as a latent prior represents a promising step toward the development of next-generation hybrid latent diffusion architectures that balance probabilistic rigor with high-speed, high-quality synthesis.

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